JENKINS 02

1 . Configure 2 slave machines in Jenkins master

- Launch an 2 ec2 instances for 2 slave machines
- Along with one master
- Install java in slave -01
- sudo yum install java-11-amazon-corretto -y

```
Installed:
  alsa-lib-1.2.7.2-1.amzn2023.0.2.x86_64
  dejavu-sans-fonts-2.37-16.amzn2023.0.2.noarch
  dejavu-serif-fonts-2.37-16.amzn2023.0.2.noarch
  fonts-filesystem-1:2.0.5-12.amzn2023.0.2.noarch
  giflib-5.2.1-9.amzn2023.0.2.x86_64
  google-noto-sans-vf-fonts-20240401-1.amzn2023.0.2.noarch
  harfbuzz-7.0.0-2.amzn2023.0.2.x86_64
  java-11-amazon-corretto-headless-1:11.0.29+7-1.amzn2023.x86_64
  langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch
  libSM-1.2.4-3.amzn2023.0.1.x86_64
  libX11-common-1.8.10-2.amzn2023.0.1.noarch
  libXext-1.3.6-1.amzn2023.0.1.x86_64
  libXinerama-1.1.5-6.amzn2023.0.1.x86_64
  libXrender-0.9.11-6.amzn2023.0.1.x86_64
  libXtst-1.2.5-1.amzn2023.0.1.x86_64
  libpng-2:1.6.37-10.amzn2023.0.6.x86_64
  pixman-0.43.4-1.amzn2023.0.4.x86_64

Complete!
[root@ip-10-0-1-87 ec2-user]# java -version
openjdk version "11.0.29" 2025-10-21 LTS
OpenJDK Runtime Environment Corretto-11.0.29.7.1 (build 11.0.29+7-LTS)
OpenJDK 64-Bit Server VM Corretto-11.0.29.7.1 (build 11.0.29+7-LTS, mixed mode)
[root@ip-10-0-1-87 ec2-user]#
```

- **On the SLAVE (Amazon Linux EC2)**
- `ssh ec2-user@SLAVE_IP`
- `ssh-keygen -t rsa -b 4096        # Press Enter for all prompts`
- `mkdir -p ~/.ssh`
- `cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys`
- `chmod 700 ~/.ssh`
- `chmod 600 ~/.ssh/authorized_keys`
- `chmod 600 ~/.ssh/id_rsa`
- Copy the private key:
- `cat ~/.ssh/id_rsa`

- **On the MASTER (Jenkins Server)**
- sudo su -
- mkdir -p /var/lib/jenkins/.ssh
- ssh-keyscan -H SLAVE_IP >> /var/lib/jenkins/.ssh/known_hosts
- chown -R jenkins:jenkins /var/lib/jenkins/.ssh
- chmod 700 /var/lib/jenkins/.ssh
- chmod 600 /var/lib/jenkins/.ssh/known_hosts

```
The authenticity of host '18.208.190.179 (18.208.190.179)' can't be establish
ED25519 key fingerprint is SHA256:XwPZQzR87/YUUw7J1KzecYm882v0KlB7Q5ud60YZ/xo
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '18.208.190.179' (ED25519) to the list of known ho
ec2-user@18.208.190.179: Permission denied (publickey,gssapi-keyex,gssapi-wit
[root@ip-10-0-1-110 .ssh]# # to become root if you must:
sudo su -
# to become ec2-user (if you're root):
su - ec2-user
Last login: Mon Nov 17 04:36:51 UTC 2025 on pts/1
[root@ip-10-0-1-110 ~]# sudo su - ec2-user
Last login: Mon Nov 17 04:27:43 UTC 2025 from 18.206.107.27 on pts/0
[ec2-user@ip-10-0-1-110 ~]$ # become root (or run with sudo)
sudo su -
mkdir -p /var/lib/jenkins/.ssh
cd /var/lib/jenkins/.ssh
# Add the slave host key to known_hosts:
ssh-keyscan -H 18.208.190.179 >> /var/lib/jenkins/.ssh/known_hosts
# Set ownership and permissions so Jenkins (user 'jenkins') can use it:
chown -R jenkins:jenkins /var/lib/jenkins/.ssh
chmod 700 /var/lib/jenkins/.ssh
chmod 600 /var/lib/jenkins/.ssh/known_hosts
Last login: Mon Nov 17 04:46:21 UTC 2025 on pts/2
```

- **i-0a4cbcc0493321880 (Jenkins)**

---**Add SSH Credential in Jenkins**

Jenkins → Manage Credentials → Global → Add Credentials

- **Kind:** SSH Username with private key
- **Username:** ec2-user
- **Private key:** paste the id_rsa from the slave
- Save

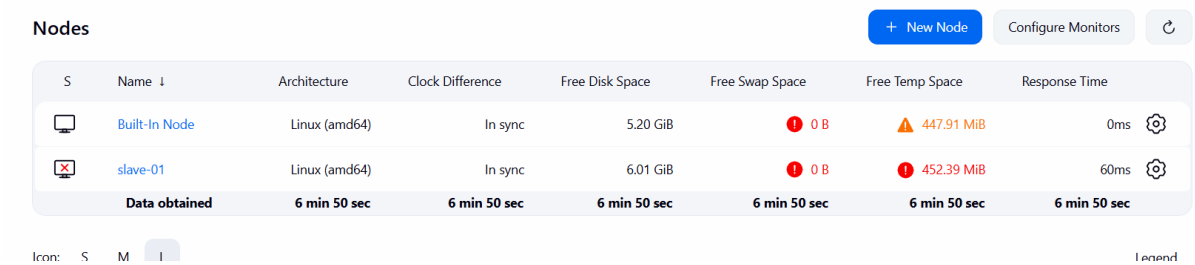
Credentials

T	P	Store ↓	Domain	ID	Name
		System	(global)	58670c12-0222-4f46-87d9-fda93cf8640f	git
		System	(global)	9cee0df0-f29b-4213-8616-8ad88e5dcc04	git-sync
		System	(global)	ec2	ec2-user (slave-01)

Create Jenkins Node

Manage Jenkins → Manage Nodes → New Node

- Name: slave-01
- Remote root dir: /home/ec2-user
- Launch method: **Launch agents via SSH**
- Host: SLAVE_IP
- Credentials: the one you added
- Host Key Strategy: **Known hosts file**
- Save → Click **Launch agent**-



- ----------

Configuring second slave -02

Repeat the same process

Create keys in slave 02

Add permission

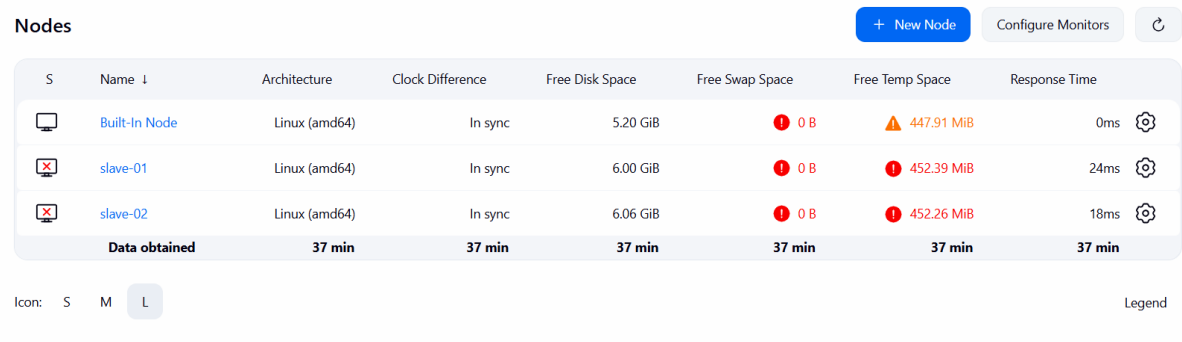
Copy private key

Add slave ip In masgter Jenkins

Now create a node in jenkins

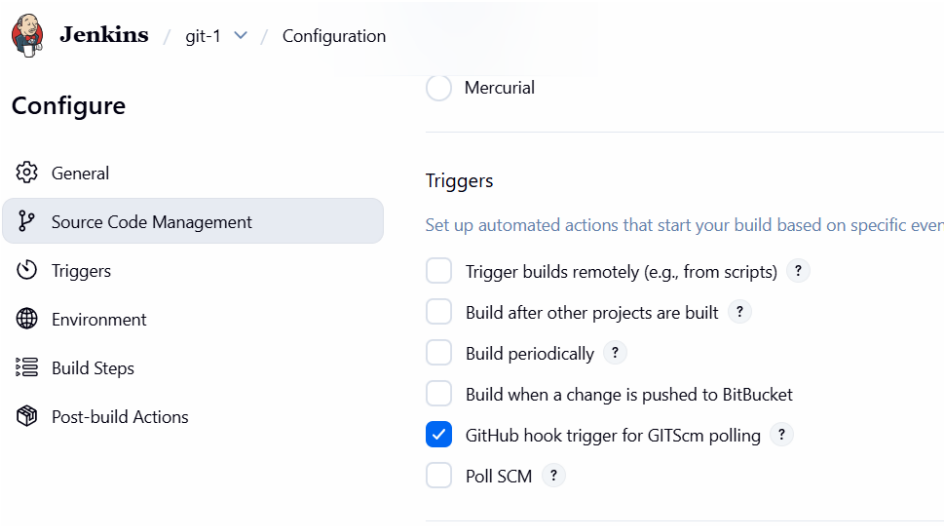
Give remote root directory

Launch the agent

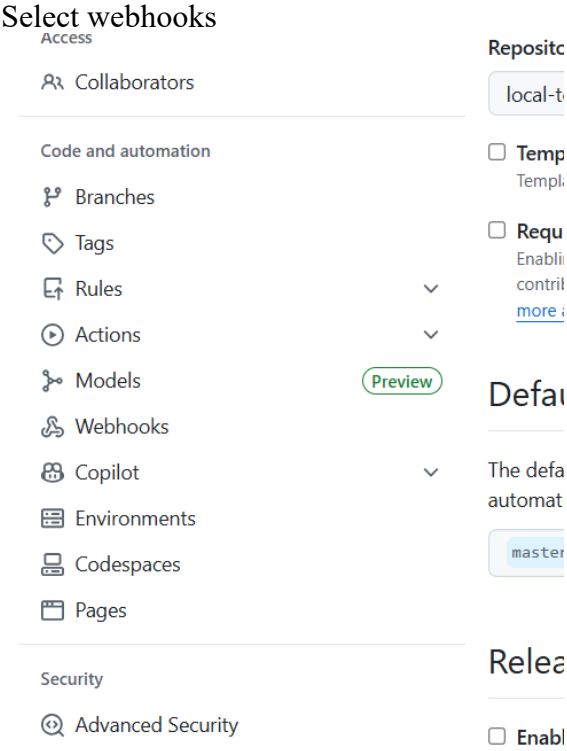


2 . Configure webhooks to Jenkins job.

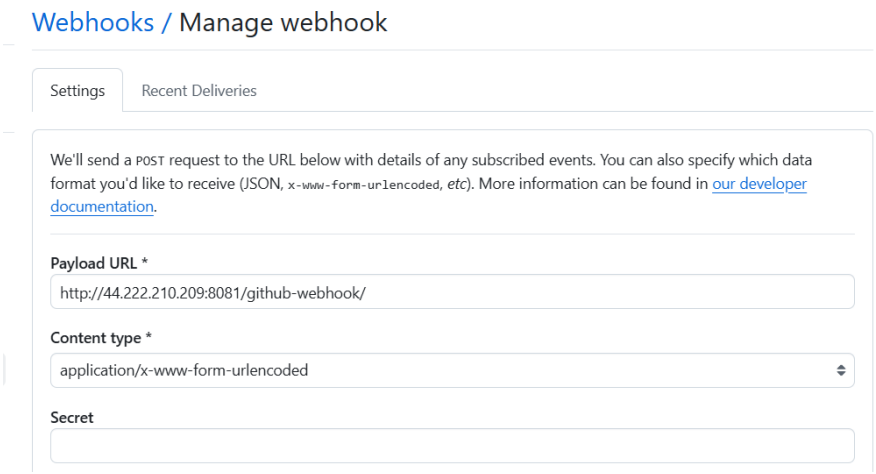
- Login to Jenkins
- Create a job
- Go to build trigger
- Select github hook trigger for GITSCM polling.



-
- Open github
- Go to repo setting
- Select webhooks



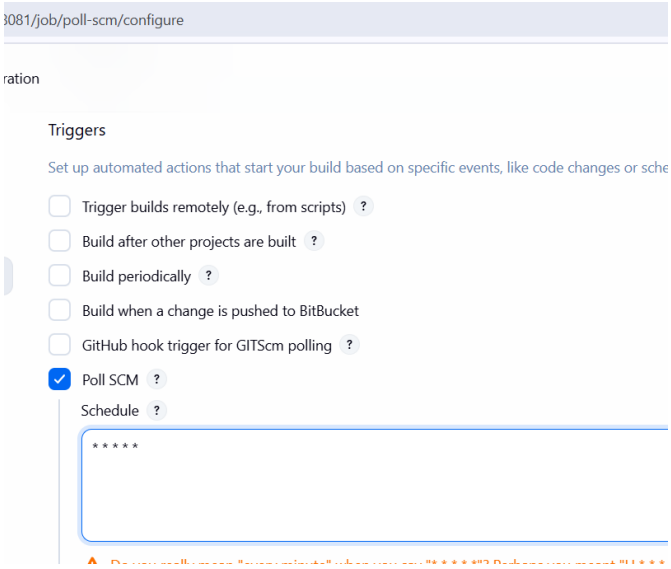
-
- Add Jenkins url and add /github-webhook/ in payload url



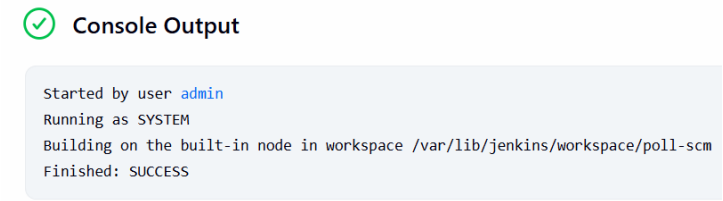
- If any changes made in repo
- Jenkins build a job

3 . Configure poll scm and build periodical options in Jenkins job.

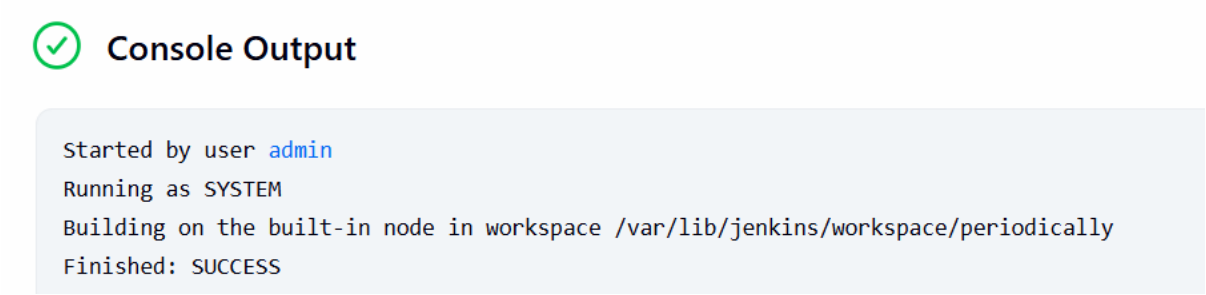
- Create a freestyle job
- Go to build trigger
- Select poll scm
- Add a cornjob
- To run job every minute . every day
- Add *****



- Apply and save it
- And build it



-
- Build periodically
- Create a free style job
- Go to build trigger
- Select build periodically
- Adda corn job for every mid night
- 0 0 * * *



4 . Take backup of Jenkins server by using bash script.

- Login to Jenkins
- Navigate to /var/lib/Jenkins/
- Ll
- Can see all the jobs, build, etc

```
[ec2-user@ip-10-0-1-110 jenkins]$ ls
com.cloudbees.hudson.plugins.folder.config.AbstractFolderConfiguration.xm
config.xml
credentials.xml
fingerprints
github-plugin-configuration.xml
hudson.model.UpdateCenter.xml
hudson.plugins.build_timeout.global.GlobalTimeOutConfiguration.xml
hudson.plugins.build_timeout.operations.BuildStepOperation.xml
hudson.plugins.emailext.ExtendedEmailPublisher.xml
hudson.plugins.git.GitSCM.xml
hudson.plugins.git.GitTool.xml
hudson.plugins.gradle.Gradle.xml
hudson.plugins.gradle.enriched.EnrichedSummaryConfig.xml
hudson.plugins.gradle.injection.InjectionConfig.xml
hudson.plugins.mercurial.MercurialInstallation.xml
hudson.plugins.timestamper.TimestamperConfig.xml
hudson.tasks.Ant.xml
hudson.tasks.Mailer.xml
hudson.tasks.Maven.xml
hudson.tasks.Shell.xml
hudson.triggers.SCMTrigger.xml
```

- Now take backup of Jenkins directory
- From root
- Sudo su
- Ls
- tar -cvf backup.tar Jenkins/
- fronm lib location can see backup file of Jenkins

```
[root@ip-10-0-1-110 lib]# ls
alternatives  dnf            initramfs  mysql      rpm        upd
amazon        games          jenkins    nfs        rpm-state  xfs
backup.tar    gssproxy       kpatch     os-prober  selinux    yum
chrony        hibinit-agent  logrotate  portables  sss
cloud         httpd          misc       private    systemd
```

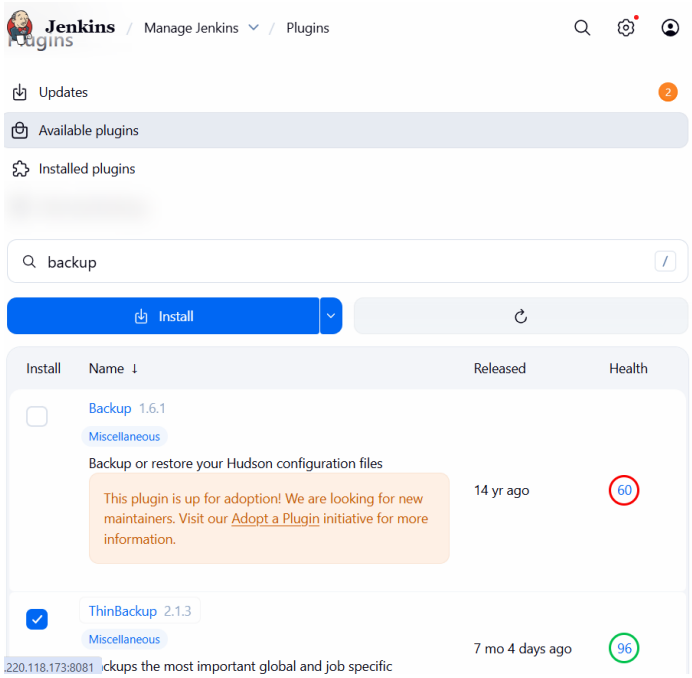
- Now create a directory
- mkdir /opt/jenkins-backups
- now add script to it fro backup
- sudo nano /opt/Jenkins-backup.sh
- add script
- #!/bin/bash
-
- BACKUP_DIR="/opt/jenkins-backups"
- JENKINS_HOME="/var/lib/jenkins"
- DATE=$(date +"%Y-%m-%d-%H-%M-%S")
- BACKUP_FILE="jenkins-backup-$DATE.tar.gz"
-
- mkdir -p $BACKUP_DIR
-
- echo "Stopping Jenkins..."
- systemctl stop jenkins
-
- echo "Creating backup..."
- tar -czvf $BACKUP_DIR/$BACKUP_FILE $JENKINS_HOME

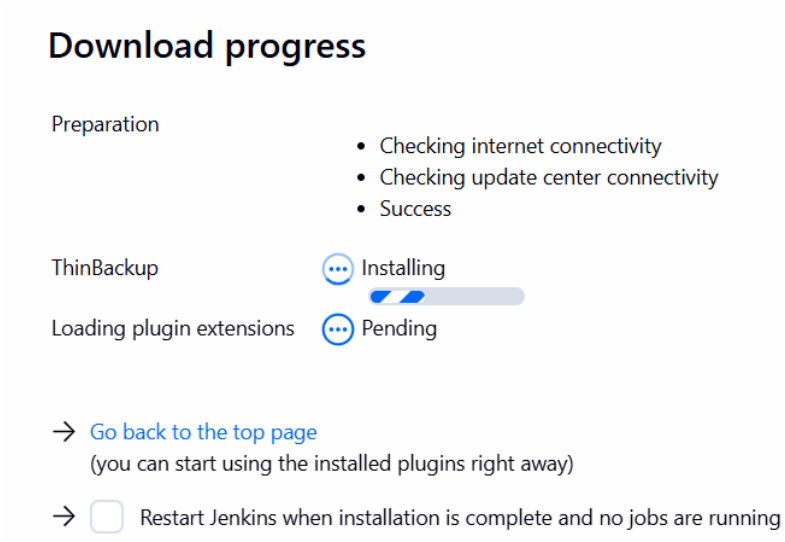
-
- echo "Starting Jenkins..."
- systemctl start jenkins
-
- echo "Backup created at: $BACKUP_DIR/$BACKUP_FILE"
- give permission to script
- Sudo chmod +x /opt/Jenkins-backup.sh
- Run the script
- /opt/jenkins-backup.sh
- Back up created scucesfully
-

```
hat-Amazon Linux-2023-en-.properties: Cannot open: Permission denied
/var/lib/jenkins/secret.key
/var/lib/jenkins/secret.key.not-so-secret
/var/lib/jenkins/plugins/
/var/lib/jenkins/plugins/commons-lang3-api.jpi
tar: /opt/jenkins-backups/jenkins-backup-2025-11-17-07-27-11.tar.gz: Canno
t write: Broken pipe
tar: Child returned status 2
tar: Error is not recoverable: exiting now
Starting Jenkins...
Failed to start jenkins.service: Access denied
See system logs and 'systemctl status jenkins.service' for details.
Backup created at: /opt/jenkins-backups/jenkins-backup-2025-11-17-07-27-11
.tar.gz
[ec2-user@ip-10-0-1-110 ~]$
```

5 .Take backup of Jenkins using rethin backup plugin.

- Login to Jenkins
- > manage Jenkins
- Go to pulgins
- Search for thin backup
- Install it
-





6 . Setup a new Jenkins server and dump the backup taken in task4.

- Launch an new instance
- In master iam having backup of Jenkins
- Now, create a file
- And add pem key to it
- Give chmod 600 permission

```
[ec2-user@ip-10-0-1-110 lib]$ ls
alternatives  backup.tar  cloud  games     hibinit-agent  initramfs
amazon        chrony      dnf    gssproxy  httpd          jenkins
[ec2-user@ip-10-0-1-110 lib]$ vi pem.key
[ec2-user@ip-10-0-1-110 lib]$ rm -rf pem.key
[ec2-user@ip-10-0-1-110 lib]$ vi test.pem
[ec2-user@ip-10-0-1-110 lib]$ sudo su
[root@ip-10-0-1-110 lib]# vi test.pem
[root@ip-10-0-1-110 lib]# chmod 600 test.pem
```

- Install Jenkins in new instance
-

```
[ec2-user@ip-172-30-0-86 ~]$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset
     Active: active (running) since Mon 2025-11-17 09:26:18 UTC; 10s ago
   Main PID: 25991 (java)
      Tasks: 44 (limit: 1053)
     Memory: 391.6M
        CPU: 21.034s
     CGroup: /system.slice/jenkins.service
             └─25991 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/

Nov 17 09:26:13 ip-172-30-0-86.ec2.internal jenkins[25991]: [LF]>
Nov 17 09:26:13 ip-172-30-0-86.ec2.internal jenkins[25991]: [LF]> **********
Nov 17 09:26:13 ip-172-30-0-86.ec2.internal jenkins[25991]: [LF]> **********
Nov 17 09:26:13 ip-172-30-0-86.ec2.internal jenkins[25991]: [LF]> **********
Nov 17 09:26:18 ip-172-30-0-86.ec2.internal jenkins[25991]: 2025-11-17 09:26
Nov 17 09:26:18 ip-172-30-0-86.ec2.internal jenkins[25991]: 2025-11-17 09:26
Nov 17 09:26:18 ip-172-30-0-86.ec2.internal systemd[1]: Started jenkins.servi
Nov 17 09:26:18 ip-172-30-0-86.ec2.internal jenkins[25991]: 2025-11-17 09:26
Nov 17 09:26:18 ip-172-30-0-86.ec2.internal jenkins[25991]: 2025-11-17 09:26
Nov 17 09:26:23 ip-172-30-0-86.ec2.internal jenkins[25991]: 2025-11-17 09:26
lines 1-20/20 (END)
```

i-0d6925402e11849f0 (backup.jenkins)

PublicIPs: 3.87.94.1 PrivateIPs: 172.30.0.86

- Now go to master and give command
- Scp -i test.pem backup.tar ec2-user@public of backup instace :/tmp
- scp -i test.pem backup.tar ec2-user@3.87.94.1:/tmp

```
[root@ip-10-0-1-110 lib]# scp -i test.pem backup.tar ec2-user@3.87.94.1:/tmp
The authenticity of host '3.87.94.1 (3.87.94.1)' can't be established.
ED25519 key fingerprint is SHA256:SxZj81G+lXsXqLZLR9BCMLxItrp1R/1CdK19JxQb1r4.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.87.94.1' (ED25519) to the list of known hosts.
backup.tar                                                    100%  372MB 185.2MB/s   00:02
[root@ip-10-0-1-110 lib]#
```

-
- Backup done
- Verify in backup instance Jenkins

```
[ec2-user@ip-172-30-0-86 ~]$ cd /tmp/
[ec2-user@ip-172-30-0-86 tmp]$ ls
backup.tar
hsperfdata_jenkins
jetty-0_0_0_0-8080-war-_-any-16829973569472035261
systemd-private-5b0a6cbd0aa84cf4a783a60bb44376a1-chronyd.service-uyinFQ
systemd-private-5b0a6cbd0aa84cf4a783a60bb44376a1-dbus-broker.service-Rso57
G
systemd-private-5b0a6cbd0aa84cf4a783a60bb44376a1-policy-routes@ens5.servic
e-2bW9WB
systemd-private-5b0a6cbd0aa84cf4a783a60bb44376a1-systemd-logind.service-lB
bvD2
systemd-private-5b0a6cbd0aa84cf4a783a60bb44376a1-systemd-resolved.service-
zO2hsd
winstone8935316054065234110.jar
[ec2-user@ip-172-30-0-86 tmp]$
```

i-0d6925402e11849f0 (backup.jenkins)

- File can seen in backup instance location cd /tmp/
- Backup done .